CSE 219 Practice Questions

Lectures 1–7: Signals, LTI Systems, Convolution, Fourier Series and Fourier Transform

Department of Computer Science and Engineering | BUET

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Lecture 1 | Introduction to Signals

Energy, power, periodicity, time transformations, and even–odd decomposition.

Problem 1.1 — Energy and average power. Consider

$$x(t) = \begin{cases} t+1, & -1 \le t < 0, \\ 2-t, & 0 \le t \le 2, \\ 0, & \text{otherwise}. \end{cases}$$

Sketch $x(t)$. Compute its total energy and its average power over the interval $[-1, 2]$.

Problem 1.2 — An infinitely long energy signal. Consider

$$x(t) = \begin{cases} 1, & |t| \le 1, \\ \dfrac{1}{|t|}, & |t| > 1. \end{cases}$$

Find its total energy and its average power over the entire time axis. (Adapted from Lecture 1.)

Problem 1.3 — Discrete-time energy. A finite-length sequence has samples

n	-2	-1	0	1	2
$x[n]$	1	-2	0	3	-1

and is zero elsewhere. Sketch the stem plot, calculate its total energy, and find its average power over $-2 \le n \le 2$.

Problem 1.4 — Two frequencies, one period. Find the fundamental period of $x(t) = \cos(4t) + 2\sin(6t)$.

Problem 1.5 — Transforming a signal. Starting from $x(t)$, write *one valid sequence* of shifts, time reversal, and time scaling that produces

$$y(t) = x\left(4 - \frac{t}{3}\right).$$

Use arrows to show the intermediate expressions.

Problem 1.6 — Even and odd components. Consider the continuous-time signal

$$x(t) = \begin{cases} t+2, & -2 \le t \le -1, \\ -t, & -1 < t \le 0, \\ 0, & \text{otherwise}. \end{cases}$$

Sketch $x(t)$ and $x(-t)$, then derive and sketch the even and odd components $x_e(t)$ and $x_o(t)$.

Problem 1.7 — Even and odd components 2.0. Consider the continuous-time signal

$$x(t) = \begin{cases} t+1, & -1 \le t \le 0, \\ 1 - \dfrac{t}{2}, & 0 < t \le 2, \\ 0, & \text{otherwise}. \end{cases}$$

Sketch $x(t)$ and $x(-t)$, then derive and sketch the even and odd components $x_e(t)$ and $x_o(t)$.

Problem 1.8 — Reversing a transformation. Suppose you are given $y(t) = x(2t-3)$, rather than $x(t)$. Describe a sequence of operations on $y(t)$ that recovers $x(t)$.

Lecture 2 | Some Useful Signals

Step and impulse functions, exponential signals, sinusoids, and discrete-time periodicity.

Problem 2.1 — A signal made of switches. Plot the discrete-time sequence

$$x[n] = 2\big(u[n+2] - u[n-1]\big) - \big(u[n-1] - u[n-3]\big).$$

Write down every nonzero sample and represent the same sequence as a sum of shifted impulses.

Problem 2.2 — The impulse picks a value. Let $x(t) = 3\delta(t+2) - 2\delta(t-1)$ and $f(t) = (t+1)^2$. Evaluate

$$\int_{-\infty}^{\infty} x(t) f(t)\, dt.$$

You may sketch the two impulses first.

Problem 2.3 — Differentiating jumps. A continuous-time signal is

$$x(t) = 2u(t+1) - 3u(t-2) + u(t-4).$$

Sketch $x(t)$ and write $\frac{d}{dt}x(t)$ as a weighted sum of impulses.

Problem 2.4 — Exponential plus a switch. Sketch $x(t) = e^{2t}u(-3t)$.

Problem 2.5 — Periodic or not. For each signal, decide whether it is periodic and, if it is, find the fundamental period:

$$\text{(a) } \cos(2t), \qquad \text{(b) } \cos(2n), \qquad \text{(c) } 3\sin\left(\frac{3\pi n}{5} + \frac{\pi}{2}\right).$$

Here (a) is continuous-time; (b) and (c) are discrete-time signals.

Problem 2.6 — Sinusoids as rotating exponentials. Express

$$x(t) = 2\sin\left(\omega_0 t + \frac{\pi}{6}\right) + \cos(2\omega_0 t)$$

as a sum of complex exponentials $c_k e^{jk\omega_0 t}$. Identify the nonzero values of c_k.

Lecture 3 | LTI Systems and Discrete Convolution

Linearity, time invariance, polynomial multiplication, impulse responses, and convolution sums.

Problem 3.1 — Convolution by sliding. Two sequences are zero outside the given ranges:

n	0	1
$x[n]$	1	2

n	0	1	2
$h[n]$	1	-1	2

Compute $y[n] = x[n] * h[n]$ using the reflect–shift–multiply–add method. Draw the output stem plot.

Problem 3.2 — Reusing a known input–output pair. For an LTI system, suppose $x_1(t) = u(t) - u(t-1)$ produces $y_1(t) = e^{-t}u(t)$. What output is produced by

$$x_2(t) = 2x_1(t+1) - x_1(t-2)?$$

.

Problem 3.3 — Build a signal out of impulses. A sequence is zero except for $x[-1] = 2$, $x[0] = -1$, and $x[2] = 3$. Write $x[n]$ as a weighted sum of shifted unit impulses. If an LTI system has impulse response $h[n]$, write its output directly in terms of shifted copies of $h[n]$.

Lecture 4 | Convolution, Causality, and Stability

Graphical convolution in discrete and continuous time; causal and stable LTI systems.

Problem 4.1 — A short graphical convolution. Compute $y[n] = x[n] * h[n]$ by drawing the overlap as $h[n-k]$ slides past $x[k]$. The nonzero samples are

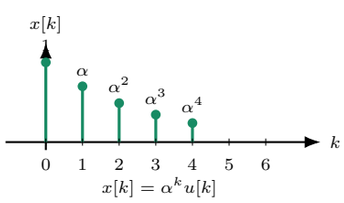
$$x[0] = \frac{1}{2}, \quad x[1] = 2, \qquad h[0] = h[1] = h[2] = 1.$$

Sketch the output, labeling every nonzero sample. (Lecture 4, Example 2.2.)

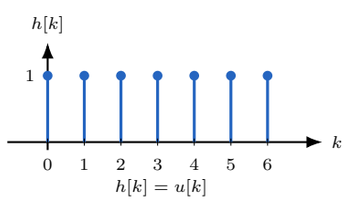
Problem 4.2 — The α convolution. Let $0 < \alpha < 1$. Compute

$$\boxed{y[n] = x[n] * h[n], \qquad x[n] = \alpha^n u[n], \qquad h[n] = u[n].}$$

Use graphical convolution: keep $x[k]$ fixed, reflect $h[k]$ to get $h[-k]$, shift to $h[n-k]$, and identify the overlap for $n < 0$ and $n \geq 0$. Obtain a closed-form formula for $y[n]$ and sketch the output.



$x[k] = \alpha^k u[k]$



$h[k] = u[k]$

Problem 4.3 — Overlapping rectangles. Compute and sketch the continuous-time convolution

$$y(t) = \underbrace{\big(u(t) - u(t-2)\big)}_{x(t)} * \underbrace{\big(u(t-1) - u(t-3)\big)}_{h(t)}.$$

Problem 4.4 — Step input, rectangular impulse response. An LTI system has $h(t) = u(t-1) - u(t-3)$. Find and sketch its response to $x(t) = u(t)$, using convolution. Label the values at $t = 1$ and $t = 3$.

Problem 4.5 — Exponential meets exponential. Find

$$y(t) = \big(e^{-2t}u(t)\big) * \big(e^{-t}u(t)\big).$$

Start by stating the overlap limits for $t < 0$ and $t \geq 0$.

Problem 4.6 — Maximum shift that preserves causality. System S_1 has impulse response $h_1(t) = u(t-1) - u(t-3)$. Another LTI system has $h_2(t) = h_1(t+a)$. Determine whether S_1 is stable, and find the *maximum* possible value of a for which S_2 is causal.

Problem 4.7 — Causal? Stable?. For each impulse response, determine whether the corresponding continuous-time LTI system is causal and whether it is BIBO-stable. Justify each conclusion using the appropriate condition:

$$\begin{aligned} h_1(t) &= e^{2t}u(-t-2), \\ h_2(t) &= te^{-t}u(t), \\ h_3(t) &= u(t), \\ h_4(t) &= 2e^{-7|t|}, \\ h_5(t) &= e^{-2t}u(-3t). \end{aligned}$$

Problem 4.8 — Periodicity versus causality and stability. Suppose a nonzero continuous function $h(t)$ is periodic with period $T > 0$, and serves as the impulse response of an LTI system. Can this system be causal? Can it be stable? Give brief mathematical reasons.

Problem 4.9 — Simulating Room Acoustics. Suppose you have a recording of someone speaking and want to simulate how their voice would sound inside a large, empty hall.

How could convolution help you achieve this? What information about the hall would you need?

Problem 4.10 — Smoothing Sensor Readings. A temperature sensor produces readings that fluctuate rapidly due to noise, even though the actual temperature changes slowly.

How could convolution be used to make the readings smoother and more reliable?

Problem 4.11 — Creating an Echo Effect. You are developing an audio editing application. A user wants to add an echo effect to a voice recording.

Explain how you could use convolution to create the effect. How could you control the delay and strength of the echo?

Lecture 5 | Fourier Series

Complex exponential coefficients, DC component, rectangular and piecewise-periodic waveforms.

Problem 5.1 — No integral needed. Write $x(t)$ in the complex exponential Fourier series form and identify all its nonzero coefficients a_k:

$$x(t) = 2 + 3\sin\left(\omega_0 t + \frac{\pi}{6}\right) + \cos(2\omega_0 t).$$

Problem 5.2 — Fourier series of a rectangular pulse train. The periodic signal $x(t)$ has period $T = 3$ and is defined over $-3/2 < t < 3/2$ by

$$x(t) = \begin{cases} 1, & -\frac{1}{2} < t < \frac{1}{2}, \\ 0, & \text{otherwise within the period.} \end{cases}$$

Find its complex Fourier coefficients a_k, handling $k = 0$ separately. Determine which nonzero harmonic indices have $a_k = 0$.

Problem 5.3 — Periodic ramp. Let $x(t) = t$ for $0 \leq t < 2$, extended periodically with $x(t+2) = x(t)$. Find the fundamental angular frequency and derive a_0 and a_k for $k \neq 0$.

Problem 5.4 — Average height of a triangular ramp. A periodic waveform with $T = 3$ is

$$x(t) = \begin{cases} t, & 0 \leq t < 1, \\ \frac{3-t}{2}, & 1 \leq t < 3, \end{cases} \qquad x(t+3) = x(t).$$

Sketch one period and calculate a_0 by geometry, without integrating. (Lecture 5's triangular-ramp waveform.)

Problem 5.5 — Fourier series: DC coefficient. Consider the continuous-time periodic signal $x(t)$ with period $T = 2$, defined over one period as

$$x(t) = \begin{cases} t^2, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2. \end{cases}$$

Let a_k be its complex exponential Fourier series coefficients. Find a_0.

Problem 5.6 — Zeros in a Fourier spectrum. Consider the continuous-time periodic signal $x(t)$ with period $T = 4$, defined over one period as

$$x(t) = \begin{cases} 1, & |t| < 1, \\ 0, & 1 < |t| < 2. \end{cases}$$

Recall that for a periodic rectangular pulse of amplitude A, width τ, and period T, centered at the origin, the complex exponential Fourier series coefficients are given by

$$a_k = \begin{cases} \dfrac{A\tau}{T}, & k = 0, \\ \dfrac{A}{k\pi} \sin\left(\dfrac{k\pi\tau}{T}\right), & k \neq 0. \end{cases}$$

Using this formula, determine the Fourier series coefficients a_k of $x(t)$ without evaluating any integrals. Also, identify the nonzero integer values of k for which $a_k = 0$.

Lecture 6 | Properties of Fourier Series

Manipulating known coefficients, differentiating jump signals, impulse trains, and coefficient convolution.

Problem 6.1 — Shift, reverse, then combine. A periodic signal $x(t)$ has period $T = 4$ and Fourier coefficients a_k. Without evaluating an integral, write the coefficients b_k of

$$y(t) = 3x(t-1) - 2x(2-t)$$

in terms of a_k, inserting the numerical value of ω_0.

Problem 6.2 — Which impulses are actually inside. Evaluate

$$\int_{-1}^{3} (t^2+1)\big[2\delta(t+2) - 3\delta(t-1) + \delta(t-2)\big]dt.$$

Show which impulses contribute, and which do not.

Problem 6.3 — Differentiation includes the jumps. A periodic signal has period $T = 2$, and one period is given by

$$x(t) = \begin{cases} 2t, & 0 \leq t < 1, \\ 0, & 1 \leq t < 2. \end{cases}$$

Sketch two periods and then sketch $\frac{d}{dt}x(t)$, including impulses at all jump locations. Write the derivative as an ordinary piecewise signal plus an impulse train.

Problem 6.4 — Fourier Series Properties. Let $x(t)$ be a differentiable periodic signal with fundamental period $T_0 = 4$ and complex exponential Fourier series coefficients a_k. Consider the signal

$$y(t) = \frac{d}{dt}\left[x(2t-1)\right].$$

Determine the fundamental period and fundamental angular frequency of $y(t)$. Express the Fourier series coefficients b_k of $y(t)$ in terms of a_k, without explicitly computing a_k.

Problem 6.5 — A train remains a train. Using the complex exponential Fourier series, prove that an impulse train in the time domain corresponds to another impulse train in the frequency domain.

Lecture 7 | Fourier Transform

From Fourier series to a continuous spectrum; rectangular transforms, translation, modulation, and impulses.

Problem 7.1 — Shifted rectangular pulse. Consider the continuous-time signal

$$x(t) = \begin{cases} 1, & 4 < t < 10, \\ 0, & \text{otherwise}. \end{cases}$$

Recall that the Fourier transform of a rectangular pulse of amplitude A and width τ, centered at the origin, is

$$X(j\omega) = \begin{cases} \dfrac{2A\sin(\omega\tau/2)}{\omega}, & \omega \neq 0, \\ A\tau, & \omega = 0. \end{cases}$$

Also, recall the time-shifting property of the Fourier transform:

$$x(t - t_0) \quad \longleftrightarrow \quad e^{-j\omega t_0} X(j\omega).$$

Using these properties, determine the Fourier transform $X(j\omega)$ of the given signal without evaluating any integrals. Also, find the value of $X(0)$.

Problem 7.2 — A Lego-block waveform. The pulse is

$$x(t) = \begin{cases} 3, & 1 < t < 2, \\ 4.5, & 2 < t < 3, \\ 3, & 3 < t < 4, \\ 0, & \text{otherwise}. \end{cases}$$

Express it as a sum of two rectangular pulses, then derive $X(j\omega)$ using only the rectangle formula and time shifting.

Problem 7.3 — Lock a cosine inside a rectangle. Let $r(t) = 1$ for $|t| < 1$ and 0 otherwise. Define

$$x(t) = r(t)\cos(3t).$$

Find $X(j\omega)$ using the frequency-shifting property. Indicate the centers of the two shifted spectral copies.

Problem 7.4 — Fourier transform of impulses. Find the Fourier transform of

$$x(t) = 3\delta(t+2) - 2\delta(t-1).$$

Also determine $X(0)$.

Problem 7.5 — Detecting Hidden Signals in Space. Astronomers receive radio signals from distant regions of space. Some signals may contain weak, repeating patterns that are difficult to notice directly.

How could the Fourier transform help astronomers detect such hidden patterns in the received signals?

Problem 7.6 — Removing Unwanted Audio Noise. A voice recording contains an unwanted high-pitched electronic tone.

How could the Fourier transform help you identify and remove this unwanted sound while preserving the speaker's voice?

Problem 7.7 — Audio Compression. Suppose you are developing an audio compression application that aims to reduce the size of audio files while preserving their perceived sound quality.

How could the Fourier transform help you achieve this? Which frequency components would you consider removing, and why?